

Zhenyu Li

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Highlights

Research Impact: [2K+](#) citations and [2K+](#) GitHub stars across open-source research projects.

Publication Record: First-author papers at top conferences (CVPR, ICCV, ECCV, ICLR), including a CVPR 2026 Highlight paper.

Industry Research Experience: [2.5+](#) years of research internships at **Meta**, **ByteDance**, **DiDi**, and **SenseTime Research**, with work spanning AR/VR, 3D reconstruction, 3D perception, autonomous driving, and foundation models.

Technical Keywords: depth estimation, 3D reconstruction, egocentric perception, generative models, diffusion models, reinforcement learning, 3D object detection, semi-supervised learning, benchmarking, VLM.

Experience

Research Intern

Meta

Zurich, Switzerland

Aug. 2025 - Feb. 2026

- Researched scalable egocentric body tracking for AR/VR, focusing on model architecture and data utilization. The work led to [EgoPoseFormer V2](#), a *CVPR 2026 Highlight* paper.
- Developed a teacher-student semi-supervised learning pipeline that scales training to tens of millions of unlabeled egocentric frames, improving accuracy while preserving low-latency inference for AR/VR deployment.

Research Intern (*TopSeed* candidate)

ByteDance

Beijing, China

Jan. 2025 - Jul. 2025

- Researched downstream benchmarking for depth foundation models, resulting in [BenchDepth](#).
- Contributed to [Depth Anything 3](#), an *ICLR 2026 Oral* paper and a state-of-the-art feed-forward 3D reconstruction system.

Elite Research Intern (Elite Camp 2022)

DiDi Cargo (ADAS team)

Beijing, China

Sep. 2022 - Mar. 2023

- Developed semi-supervised learning methods for monocular 3D object detection, improving performance under limited labeled data. The work led to [3DSeMo](#), an *ICRA 2025* paper.

Research Intern

SenseTime Research (ADAS team)

Shanghai, China

Jan. 2022 - Jul. 2022

- Developed unsupervised domain adaptation methods for monocular 3D object detection, leading to [STMono3D](#), an *ECCV 2024* paper.
- Deployed the domain adaptation pipeline in an industrial collaboration with GAC Group and met project performance targets.
- Extended the work to domain generalization for monocular 3D object detection, leading to [DGMono3D](#).
- Contributed to 3D object detection research that led to [BEVDistill](#), an *ICLR 2023* paper.

Perception Algorithm Development Intern

SenseTime Research (ADAS team)

Shanghai, China

Mar. 2021 - Sep. 2021

- Built a ReID dataset from a ground-truth annotation system, trained a ReID model, and integrated it into an ADAS perception pipeline.
- Implemented and deployed a C++ multi-object tracking module for ADAS, incorporating appearance features and cascade association.
- Researched multimodal contrastive learning for spatially aware visual representations, leading to [SimIPU](#), an *AAAI 2022* paper.
- Contributed to 3D object detection research that led to [AutoAlign](#) and [AutoAlignV2](#), published at *IJCAI 2022* and *ECCV 2022*, respectively.

Selected First Author Publications

- **EgoPoseFormer V2: Accurate Egocentric Human Motion Estimation for AR/VR.** *CVPR 2026 Highlight*. Zhenyu Li, Sai Kumar Dwivedi, Filip Maric, Carlos Chacón, Nadine Bertsch, Filippo Arcadu, Tomas Hodan, Michael Ramamonjisoa, Peter Wonka, Amy Zhao, Robin Kips, Cem Keskin, Anastasia Tkach, Chenhongyi Yang. [Project link](#).
- **PatchRefiner V2: Fast and Lightweight Real-Domain High-Resolution Metric Depth Estimation.** *ICLR 2026*. Zhenyu Li, Wenqing Cui, Shariq Farooq Bhat, Peter Wonka. [Project link](#).
- **Amodal Depth Anything: Amodal Depth Estimation in the Wild.** *ICCV 2025*. Zhenyu Li, Mykola Lavreniuk, Jian Shi, Shariq Farooq Bhat, Peter Wonka. [Project link](#).
- **PatchRefiner: Leveraging Synthetic Data for Real-Domain High-Resolution Monocular Metric Depth Estimation.** *ECCV 2024*. Zhenyu Li, Shariq Farooq Bhat, Peter Wonka. [Project link](#).
- **PatchFusion: An End-to-End Tile-Based Framework for High-Resolution Monocular Metric Depth Estimation.** *CVPR 2024*. Zhenyu Li, Shariq Farooq Bhat, Peter Wonka. [Project link](#).
- **SimIPU: Simple 2D Image and 3D Point Cloud Unsupervised Pre-Training for Spatial-Aware Visual Representations.** *AAAI 2022*. Zhenyu Li, Zehui Chen, Ang Li, Liangji Fang, Qinrong Jiang, Xianming Liu, Junjun Jiang, Bolei Zhou, Hang Zhao. [Project link](#).
- **BinsFormer: Revisiting Adaptive Bins for Monocular Depth Estimation.** *Transactions on Image Processing*. Zhenyu Li, Xuyang Wang, Xianming Liu, Junjun Jiang. [Project link](#).
- **CriterAlign: Criterion-Centric Rationale Alignment for Code Preference Judging.** *Under Review*. Zhenyu Li, Aleksandar Cvejic, Zehui Chen, Peter Wonka. [Project link](#).

Selected Co-Author Publications

- **Depth Anything 3: Recovering the Visual Space from Any Views.** *ICLR 2026 Oral*. Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, Bingyi Kang. [Project link](#).
- **DivRL: Disentangled Self-Similarity Rewards for Diverse Subject-Driven Generation.** *ECCV 2026*. Qian Wang, Zhenyu Li, Abdelrahman Eldesokey, Peter Wonka.
- **Any Resolution Any Geometry: From Multi-View To Multi-Patch.** *CVPR 2026*. Wenqing Cui, Zhenyu Li, Mykola Lavreniuk, Jian Shi, Ramzi Idoughi, Xiangjun Tang, Peter Wonka. [Project link](#).
- **DissolveStereo: Coarse Depth Injection for Zero-Shot Stereo Video Generation.** *SIGGRAPH 2026*. Jian Shi, Qian Wang, Zhenyu Li, Wenqing Cui, Ramzi Idoughi, Peter Wonka. [Project link](#).
- **ImmersePro: End-to-End Stereo Video Synthesis via Implicit Disparity Learning.** *ICML 2026*. Jian Shi, Zhenyu Li, Peter Wonka. [Project link](#).
- **Lari: Layered Ray Intersections for Single-View 3D Geometric Reasoning.** *ICML 2026*. Rui Li, Biao Zhang, Zhenyu Li, Federico Tombari, Peter Wonka. [Project link](#).
- **Artic-O: End-to-End Articulated Object Reconstruction via Latent Geometry Learning.** Xuyang Wang, Zhenyu Li, Jian Ding, Habib Slim, Peter Wonka, Hongdong Li, Mohamed Elhoseiny.

Education

King Abdullah University of Science and Technology (KAUST) <i>PhD in Computer Science, Advisor: Prof. Peter Wonka</i>	Thuwal, Saudi Arabia <i>Sep. 2023 - Present</i>
Harbin Institute of Technology (HIT) <i>MS in Computer Science, Advisor: Prof. Junjun Jiang. Program rank - 2/80+</i>	Harbin, China <i>Sep. 2021 - Jul. 2023</i>
Harbin Institute of Technology (HIT) <i>BS in Computer Science, Advisor: Prof. Junjun Jiang. GPA - 89.58/100</i>	Harbin, China <i>Sep. 2017 - Jul. 2021</i>

Awards

Dean's List Award, KAUST CEMSE , awarded to top students for academic achievement	2024
ICCV VCL 2023 Multitask Learning for Robustness Challenge, 1st place	2023
ECCV SSLAD 2022 3D Object Detection Challenge, 3rd Place	2022
ECCV Mobile AI & AIM 2022 Monocular Depth Estimation Challenge, 2nd Place	2022
China National Scholarship	2022